

The Haigh–Mallion Ring-Current Surface Calculator

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1 What the calculator computes

The Haigh–Mallion calculator represents the magnetic shielding pattern caused by an aromatic ring current. When an external field is applied to an aromatic ring, the delocalized π electrons can circulate around the ring. That circulation produces a secondary magnetic field at a nearby nucleus. The secondary field changes the nuclear shielding tensor σ , and the isotropic part of σ contributes to the chemical shift reported in solution NMR.

This implementation does not solve the electronic response of the ring. It takes the ring geometry from the protein structure, integrates a magnetic dipole kernel over the polygonal ring surface, and stores the resulting geometric tensor. The output has units $\text{\AA}^{-1}$. It is not a reported shielding value in ppm and does not stand on its own as a chemical shift. After calibration against DFT shielding tensors, it can contribute a ring-current shielding term; before calibration, it is a geometry-derived descriptor whose tensor shape is the object of interest.

2 Where shielding enters

For a nucleus in an applied magnetic field B_0 , the field experienced by the nucleus is written

$$B_{\text{nuc}} = (I - \sigma)B_0. \quad (1)$$

Here I is the 3×3 identity matrix. The shielding tensor σ is dimensionless in physical NMR: it is the linear-response coefficient connecting the applied field to the electronic secondary field at the nucleus. It is a matrix because the molecule has orientation. A field applied along direction b can induce a secondary-field component along direction a , so σ_{ab} records how those two directions are coupled.

The quantum-mechanical content behind Eq. (1) is that the external magnetic field perturbs the electronic state. The perturbed electrons generate a magnetic field of their own, and the nucleus responds to the sum. The Haigh–Mallion calculator keeps one classical, geometry-based piece of that response. For an aromatic ring, the component of B_0 normal to the ring threads magnetic flux through the π -electron system. The model represents the induced current response by a thin magnetic surface lying inside the ring.

The sign convention in the code is

$$\sigma_{ab} = -\frac{\partial B_a^{\text{sec}}}{\partial B_{0,b}}. \quad (2)$$

The minus sign says that shielding is the reduction of the applied field at the nucleus. In the source, this sign appears when the full kernel G_{ab} is formed from the surface integral. A diamagnetic literature ring-current intensity is negative, but that intensity is not multiplied into this calculator's stored result.

3 The tensor split used by the code

Any 3×3 tensor X is decomposed by `SphericalTensor::Decompose` before it is emitted. The scalar part is

$$T_0 = \frac{1}{3} \text{tr } X. \quad (3)$$

This is the isotropic component: it multiplies I , survives molecular tumbling, and sets the isotropic shielding value before reference subtraction.

The antisymmetric part is packed as the three-component vector

$$\mathbf{T}_1 = \frac{1}{2} \begin{pmatrix} X_{yz} - X_{zy} \\ X_{zx} - X_{xz} \\ X_{xy} - X_{yx} \end{pmatrix}. \quad (4)$$

This vector is the dual form of the part of $\mathbf{X}$ that changes sign under transpose. It is usually not observed directly in standard isotropic solution NMR, but it is retained because the Haigh–Mallion shielding kernel need not be symmetric.

The remaining symmetric traceless matrix is

$$\mathbf{S} = \frac{\mathbf{X} + \mathbf{X}^\top}{2} - T_0 \mathbf{I}, \quad \text{tr } \mathbf{S} = 0. \quad (5)$$

It has five independent entries. The code stores those entries in the real-spherical T_2 basis

$$\mathbf{T}_2 = \begin{pmatrix} \sqrt{2} S_{xy} \\ \sqrt{2} S_{yz} \\ \sqrt{3/2} S_{zz} \\ \sqrt{2} S_{xz} \\ (S_{xx} - S_{yy})/\sqrt{2} \end{pmatrix}. \quad (6)$$

These factors are the normalization used by the implementation. They preserve the Frobenius norm, so the squared length of the five-vector equals $\sum_{ab} S_{ab}^2$. The nine stored values are ordered as

$$[T_0, T_{1x}, T_{1y}, T_{1z}, T_{2,-2}, T_{2,-1}, T_{2,0}, T_{2,+1}, T_{2,+2}].$$

The Haigh–Mallion path creates two tensors. The raw surface integral $\mathbf{H}$ is symmetric and traceless up to floating-point quadrature error, so its decomposition is carried by $\mathbf{T}_2$. The stored shielding kernel $\mathbf{G}$, formed after contraction with the ring normal, is a rank-one matrix and can have nonzero T_0 , T_1 , and T_2 . The per-atom output uses the decomposition of the accumulated $\mathbf{G}$, not the raw $\mathbf{H}$.

4 The method implemented

For each protein conformation, the calculator depends on the precomputed ring geometry and a spatial index. The ring list is the aromatic part of the protein topology: PHE benzene, TYR phenol, TRP benzene, TRP pyrrole, TRP indole perimeter, HIS imidazole, HID imidazole, and HIE imidazole. A saturated PRO pyrrolidine type exists elsewhere in the topology, but the aromatic ring-count API used here does not present it to this loop.

For a ring with vertices $\mathbf{q}_i$, the geometry routine sets the center to the average vertex position,

$$\mathbf{c} = \frac{1}{N} \sum_{i=1}^N \mathbf{q}_i, \quad (7)$$

with positions measured in Å. The ring normal $\hat{\mathbf{n}}$ is the normal to the best-fit plane from an SVD of the centered vertex coordinates. Its sign is chosen to agree with $(\mathbf{q}_1 - \mathbf{q}_0) \times (\mathbf{q}_2 - \mathbf{q}_0)$. The radius used by filters is the average distance of the vertices from $\mathbf{c}$.

For a target atom at position $\mathbf{r}$, the raw tensor is

$$H_{ab}(\mathbf{r}) = \int_S \left(\frac{3\rho_a\rho_b}{\rho^5} - \frac{\delta_{ab}}{\rho^3} \right) dS, \quad \boldsymbol{\rho} = \mathbf{r} - \mathbf{s}, \quad \rho = |\boldsymbol{\rho}|. \quad (8)$$

The integration point $\mathbf{s}$ runs over the ring surface S ; dS is an area element in Å²; δ_{ab} is one when $a = b$ and zero otherwise. The integrand has units Å⁻³, so $\mathbf{H}$ has units Å⁻¹. The matrix inside the integral is

the Cartesian magnetic-dipole field kernel: if a unit point dipole at s points along direction b , the expression gives the field component along direction a at r , apart from physical constants not stored here. The Haigh–Mallion surface is therefore a sheet of infinitesimal dipoles. Integrating Eq. (8) asks what field that sheet would produce at the nucleus for each possible dipole direction.

The ring current is tied to the ring normal. The implementation contracts $\mathbf{H}$ with $\hat{\mathbf{n}}$,

$$\mathbf{V}(\mathbf{r}) = \mathbf{H}(\mathbf{r})\hat{\mathbf{n}}, \quad (9)$$

so V_a is the effective secondary-field component from a unit surface response normal to the ring. The applied field enters through its normal component, $\hat{\mathbf{n}} \cdot \mathbf{B}_0 = n_b B_{0,b}$. Combining that flux factor with the shielding sign convention gives the stored tensor

$$G_{ab}(\mathbf{r}) = -V_a(\mathbf{r})n_b = -[\mathbf{H}(\mathbf{r})\hat{\mathbf{n}}]_a n_b. \quad (10)$$

This is an outer product, so the Cartesian matrix has rank one unless $\mathbf{V} = \mathbf{0}$. Its units remain $\text{\AA}^{-1}$, because $\hat{\mathbf{n}}$ is dimensionless.

The surface S is not meshed by an external library. The code creates a fan triangulation from the ring center to each adjacent vertex pair. On each triangle $T = (v_0, v_1, v_2)$, it evaluates

$$\int_T F(\mathbf{s}) dS \approx A_T \sum_{q=1}^7 w_q F(\lambda_{q0}\mathbf{v}_0 + \lambda_{q1}\mathbf{v}_1 + \lambda_{q2}\mathbf{v}_2), \quad (11)$$

where A_T is the triangle area. The λ 's are barycentric coordinates: nonnegative weights that sum to one and locate a point inside the triangle. The seven quadrature weights also sum to one. The rule is the degree-five Stroud T2:5-1, also listed as Dunavant's seven-point triangle rule; the code computes its two barycentric orbit parameters $(6 \pm \sqrt{15})/21$ and weights $9/40$ and $(155 \pm \sqrt{15})/1200$.

Near the ring plane, the ρ^{-3} kernel changes rapidly. The implementation refines a fan triangle by splitting it at edge midpoints when the target atom is close to a triangle vertex. The first split is triggered below $2.0 \text{ \AA}$, the second below $1.0 \text{ \AA}$, and the depth stops at two. That leaves 7 quadrature points without subdivision, 28 after the first split, and at most 112 if all level-1 subtriangles split. A triangle whose area is below $10^{-20} \text{ \AA}^2$ is skipped, and a quadrature point closer than $0.1 \text{ \AA}$ to the target atom is skipped.

Candidate ring centers are first found within $15.0 \text{ \AA}$ of the target atom. Three filters then decide whether the atom–ring pair is evaluated. The $0.1 \text{ \AA}$ singularity guard rejects a target closer than that distance to the expansion center. A near-field filter rejects points whose center distance is not larger than half the source extent; for rings the source extent is the diameter, so the threshold is the ring radius. A topology filter rejects ring vertices and atoms covalently bonded to ring vertices, because those atoms lie in a through-bond regime rather than the through-space surface model.

The TOML file also contains ring-current intensities in nA/T: PHE -12.0 , TYR -11.28 , TRP6 -12.48 , TRP5 -6.72 , HIS/HID/HIE -5.16 , and TRP9 -19.2 . It contains Johnson–Bovey lobe offsets in $\text{\AA}$: 0.64 for PHE, TYR, and TRP6; 0.52 for TRP5; 0.50 for HIS/HID/HIE; and 0.60 for TRP9. This calculator reads neither set. The source computes G from geometry, records the ring type for per-type sums, and lets calibration or a later model decide how strongly each ring type should contribute.

5 Inputs and output

The direct inputs are atomic coordinates, typed aromatic-ring topology, the ring geometry derived from those coordinates, and the spatial index used to find nearby rings. The calculator does not consume MOPAC charges or bond orders, AIMNet2 charges or embeddings, ff14SB charges, APBS electrostatic fields, or DSSP secondary-structure assignments. Those data sources feed other calculators or provide structural context; their internal methods are outside the scope of this document.

For each atom, the calculator stores per-ring attribution: $\mathbf{H}$, its decomposition, $\mathbf{V} = \mathbf{H}\hat{\mathbf{n}}$, G , and the decomposition of G . It accumulates the accepted G tensors over nearby rings and writes `hm_shielding.npy`

with shape $N \times 9$, using the $[T_0, T_1, T_2]$ order given after Eq. (6). It also writes `hm_per_type_T0.npy` with shape $N \times 8$ and `hm_per_type_T2.npy` with shape $N \times 40$, one T_0 scalar and five T_2 components for each aromatic ring type.

The grid-sampling method evaluates the same G construction at an arbitrary point. It does not use the bonded topology filter, and it skips rings closer than the singularity guard, points inside the ring radius, and rings beyond 15.0 Å.

Despite the output filename, the values are geometric kernels. Their role in the thesis is to test whether the Haigh–Mallion geometry carries the tensor signal seen in DFT shielding differences. Agreement is therefore assessed as correlation after calibration, especially in the T_2 anisotropic components, not as pointwise reproduction of a chemical shift.